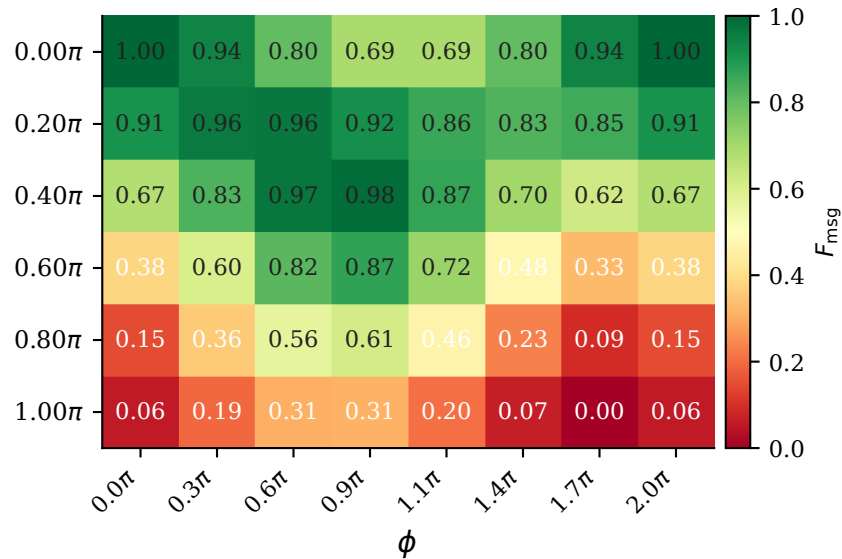
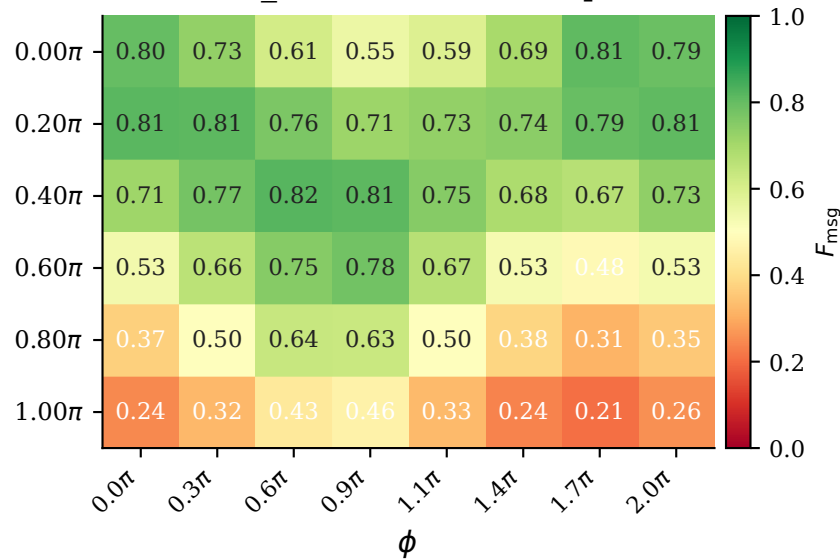


Parameter sweep: $F_{\text{msg}}(\theta, \phi)$ — ibm_torino vs noiseless simulator
 Grid: 6×8, Spearman $r = 0.9497$ ($p = 7.9e - 25$), Pearson $r = 0.9709$

(a) Simulator $F_{\text{msg}}(\theta, \phi)$
 (noiseless)



(b) Hardware $F_{\text{msg}}(\theta, \phi)$
 (ibm_torino, 2000 shots/pt)



(c) $\Delta F = F_{\text{hw}} - F_{\text{sim}}$
 mean=+0.000, std=0.140

